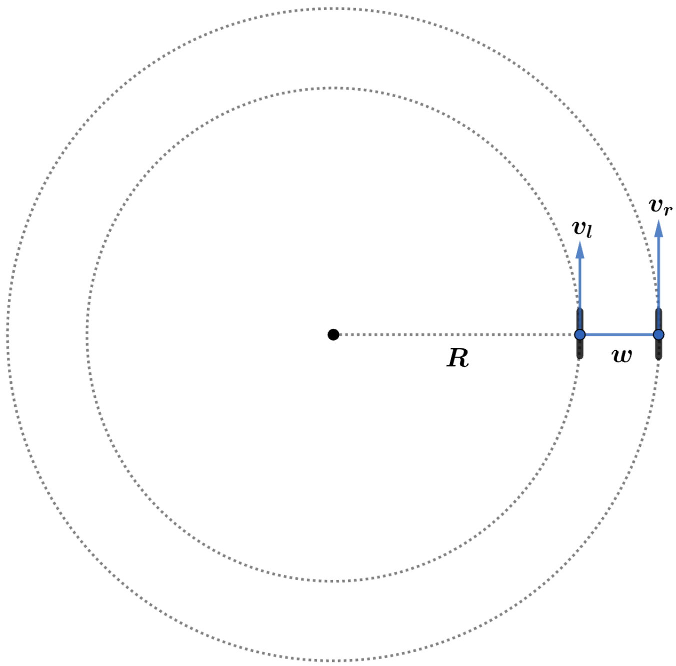
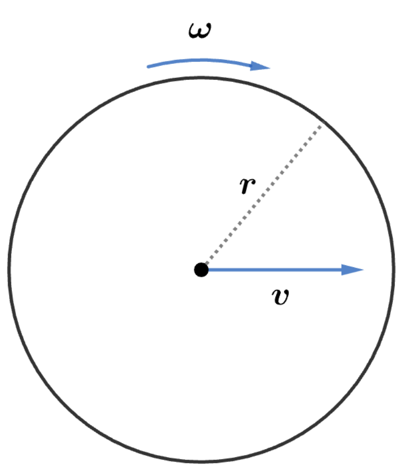
2016 F=ma Exam: Problem 1

Kevin S. Huang



Since the car (rigid object) is traveling in a circle, all parts of the car move at the same angular velocity ω_{car} around the center. If the left (right) wheel has velocity v_l (v_r) then we must have

$$\omega_{\text{car}} = \frac{v_l}{R} = \frac{v_r}{R+w}$$
$$\frac{v_l}{v_r} = \frac{R}{R+w}$$



The velocity v of the wheel is related to the angular velocity ω around its axle by

$$v = r\omega$$

since the wheels roll without slipping. Thus,

$$\frac{\omega_l}{\omega_r} = \frac{v_l}{v_r} = \frac{R}{R+w} = \frac{9.60\,\mathrm{m}}{9.60\,\mathrm{m} + 1.74\,\mathrm{m}} = 0.847$$

so the answer is $\boxed{\mathrm{E}}$.